PART -B

1. Explain in details the steps involved in the data science process, from defining research goals to presenting findings and building applications

The data science process is a systematic approach for extracting insights from data, and it involves several key stages, from problem definition to application deployment. Data science uses different methods and concepts, including statistics, to make sense of large datasets and inform decision-making. Data warehousing and data mining are two complementary techniques that play distinct roles in this process.

The data science process is an iterative workflow that turns raw data into actionable knowledge. The following are the key steps involved:

1. Business understanding and problem definition

Define research goals: The first step is to clearly define the problem and objectives of the project. This is done by engaging with stakeholders to understand the business need, such as reducing customer churn, predicting sales, or optimizing a logistics route.

Project charter: A project charter formalizes the goals, deliverables, resource requirements, and timelines to ensure everyone is aligned.

2. Data retrieval

Data collection: Gather relevant data from various internal and external sources. These can include databases (SQL, NoSQL), APIs from web services, web scraping for public data, and real-time streams from IoT sensors.

Data quality checks: Perform initial checks during retrieval to identify data type errors, typos, and other simple inconsistencies before deeper analysis.

3. Data preparation

Data cleansing: Cleanse the raw data by handling missing values (e.g., imputing or removing), removing duplicate entries, and correcting inconsistencies (e.2.3).

Data integration: Combine data from different sources into a unified, consistent format, which is a key function of data warehousing.

4. Data transformation and feature engineering: Transform and scale the data to make it suitable for modeling. This can include creating new, more meaningful features from existing ones to improve model performance. For example, a customer's birthdate could be used to create a new "age" feature.

Compiled by Manikandan R

5. Exploratory data analysis (EDA)

Summarize data: Use statistical measures to understand the data's characteristics, such as central tendency (mean, median, mode) and dispersion (standard deviation, range).

Visualize data: Create plots and charts (e.g., histograms, scatter plots) to uncover patterns, trends, relationships, and anomalies. This step guides the modeling approach.

6. Data modeling

Select and build models: Apply machine learning algorithms and statistical techniques to the prepared data. This could involve supervised learning for prediction (e.g., classification, regression), unsupervised learning for finding hidden structures (e.g., clustering), or other advanced techniques.

Training and testing: Split the data into training and testing sets to train the model and evaluate its performance on unseen data.

7. Model evaluation and optimization

Validate performance: Assess the model's performance using appropriate metrics (e.g., accuracy, precision, recall).

Refine the model: Optimize the model's parameters and repeat the process if performance is not satisfactory. This is often an iterative process that may involve looping back to earlier stages.

8. Presenting findings and deploying applications

Communicate insights: Present the results and insights to stakeholders using clear language and effective visualizations. Storytelling with data is crucial for demonstrating business value.

Model deployment: Integrate the finalized model into a real-world production environment. This can be a web application, a mobile app, or an automated process.**Monitoring and maintenance:** Continuously monitor the model's performance to ensure it remains accurate and relevant as conditions change over time.

2. **Explain the data science process with a diagram**

The data science process is a cyclical workflow involving problem definition, data collection, data preparation (cleaning and transformation), exploratory data analysis, model building and evaluation, and deployment and communication of results. It's iterative, meaning teams may loop back to earlier stages to refine goals, improve data, or adjust models as they gain insights.

Here's a detailed breakdown of the steps:

1. Business Understanding & Problem Definition

Goal:

Compiled by Manikandan R

To clearly understand the business problem, the objectives, and how data science can help solve it.

Tasks:
Define the scope of the project, identify the questions to be answered, and establish clear, measurable goals.

2. Data Collection

Goal: Gather all relevant raw data from various sources.

Tasks: Data acquisition from databases, APIs, web scraping, surveys, or other systems.

3. Data Preparation (Data Cleaning & Feature Engineering)

Goal: Transform raw data into a usable format for analysis and modeling.

Tasks:
Data Cleaning: Handle missing values, correct inaccuracies, and remove duplicate records.

Data Transformation: Standardize formats, integrate data from different sources, and aggregate data.

Feature Engineering: Create new variables or select existing ones that will be most effective for modeling.

4. Data Exploration & Analysis (EDA)

Goal:

Understand the data's "look and feel," find patterns, trends, and anomalies.

Tasks:

Use visualizations like histograms, scatter plots, and box plots to gain initial insights. Perform statistical analysis, clustering, and classification to understand data distributions and identify potential biases.

5. Model Building & Algorithm Selection

Goal: Develop predictive or analytical models to address the project goals.

Tasks:
Algorithm Selection: Choose appropriate machine learning or statistical algorithms based on the problem type (e.g., regression, classification) and data characteristics.

Model Training: Use the prepared data to train the selected models.

6. Model Evaluation

- **Goal:** Assess the model's performance and ensure it meets the defined goals.
- **Tasks:** Validate the model using appropriate metrics to determine its accuracy and reliability.

7. Deployment & Communication

- **Goal:**

Implement the model into a production system and communicate findings to stakeholders.

Compiled by Manikandan R

- **Tasks:**
- **Deployment:** Integrate the model into a production environment or a data pipeline for continuous use.
- **Communication:** Present insights, findings, and recommendations to decision-makers using data visualizations, reports, and dashboards to facilitate understanding and action.

8. Iteration & Monitoring

- **Goal:**

  Continuously refine and update the model and process based on new data or performance.

- **Tasks:**
  Monitor model performance in production, collect feedback, and loop back to earlier stages if necessary to refine the problem, data, or models.

3. **Compare and contrast data mining and data warehousing , highlighting their respective roles in data science.**

- Data mining and data warehousing are two distinct yet complementary fields within data science. A data warehouse provides the clean, organized foundation that data mining tools need to operate effectively.

Aspect	Data Warehousing	Data Mining
Purpose	To store and organize large volumes of integrated, structured, and historical data from various sources.	To analyze data and discover hidden patterns, trends, correlations, and anomalies within large datasets.
Process	Involves the **ETL (Extract, Transform, Load)** process, where data is extracted from source systems, transformed into a consistent format, and loaded into the warehouse.	Uses algorithms and statistical techniques, such as classification, clustering, and association rule mining, to analyze data.
Functionality	Supports business intelligence (BI) tools for querying and reporting. It provides a single source of truth for historical data analysis.	Enables predictive and descriptive analytics by extracting deeper, more complex insights from the data stored in the warehouse.
Data State	Holds structured data that is subject-oriented, integrated, time-variant, and non-volatile.	Works with both structured and unstructured data, using the warehouse's organized data as its primary input.
Users	Primarily used by data engineers, BI teams, and management for routine	Used by data scientists, machine learning engineers,

Compiled by Manikandan R

	reporting and decision-making.	and analysts to uncover more complex insights.
Analogy	A library that collects, organizes, and stores books for easy retrieval.	A researcher who studies the books in the library to find new knowledge and connections.
Role in Data Science	Acts as the **foundation** for data analysis by providing a structured and reliable data source.	Represents the **analytical core**, where models are built to generate actionable intelligence and predictions.

4. Discuss the benefits and applications of data science.

Data science has transformed industries by turning raw data into valuable, actionable insights.

Benefits of data science

Improved decision-making: Data science provides objective evidence and insights, enabling companies to make informed, data-driven decisions rather than relying on guesswork.

Enhanced performance measurement: By analyzing KPIs and other metrics, organizations can more accurately track performance, identify inefficiencies, and optimize operations.

Risk mitigation and fraud detection: Machine learning models can analyze transaction patterns to detect anomalies, helping financial institutions and other businesses identify and prevent fraud.

Increased efficiency: Data science helps streamline processes, automate repetitive tasks, and optimize resource allocation, leading to cost savings and higher productivity.

Improved customer experience: By analyzing customer data, companies can offer personalized recommendations, tailor products, and enhance customer satisfaction and loyalty.

Predictive insights: Using predictive analytics, businesses can forecast future trends, market conditions, and consumer behavior, giving them a competitive edge.

Product innovation: Data-driven insights into customer behavior and preferences allow companies to develop better, more user-centric products.

- **Applications of data science**

Healthcare: Data science is used for predictive analytics to forecast disease outbreaks, for personalized medicine based on genetic data, and for medical image analysis.

Compiled by Mahendran R

E-commerce: Companies like Amazon use data science for personalized product recommendations, customer segmentation for targeted marketing, and optimized inventory management.

Finance: Applications include credit risk assessment, algorithmic trading, and robust fraud detection systems.

Transportation and logistics: Data science helps in optimizing delivery routes, predicting equipment failure for preventative maintenance, and managing traffic flow.

Manufacturing: Predictive maintenance uses data from sensors to forecast equipment failures and minimize costly downtime.

Marketing: Data science enables effective customer segmentation and targeted marketing campaigns to maximize ROI.

Tech (AI/ML): Data science is the foundation for speech recognition (virtual assistants), image recognition (face detection), and building recommendation engines.

5. Explain basic statistical descriptions of data, including central tendency and dispersion.

Descriptive statistics are used to summarize and organize data in a meaningful way. The two main types are measures of central tendency and measures of dispersion.

- **Measures of central tendency**

These measures identify the center or typical value of a dataset. Key measures include:
Mean: The average of all values, calculated by summing all values and dividing by the count. An example for the dataset {2, 3, 5, 8, 12} is 6.

Mean Formula

$$mean = \frac{\text{sum of the terms}}{\text{number of terms}}$$

Median: The middle value of a sorted dataset, less affected by extreme values than the mean. For {2, 3, 5, 8, 12}, the median is 5.

Median Formula

if n is odd,

$$median = \left(\frac{n+1}{2}\right)^{th}$$

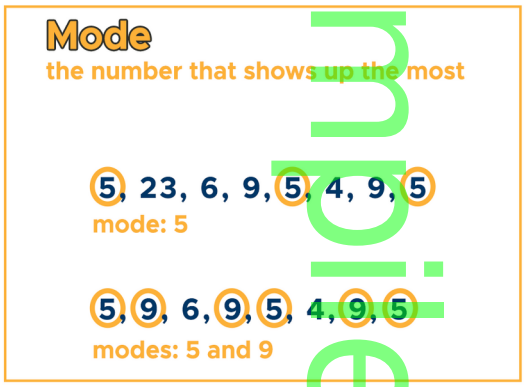
if n is even,

$$median = \frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2}+1\right)^{th}}{2}$$

n = number of terms
th = n(th) number

Prepared by Manpreet Kaur

Mode: The most frequent value in a dataset, applicable to both numerical and categorical data. In {2, 3, 3, 5, 8, 8, 8}, the mode is 8.



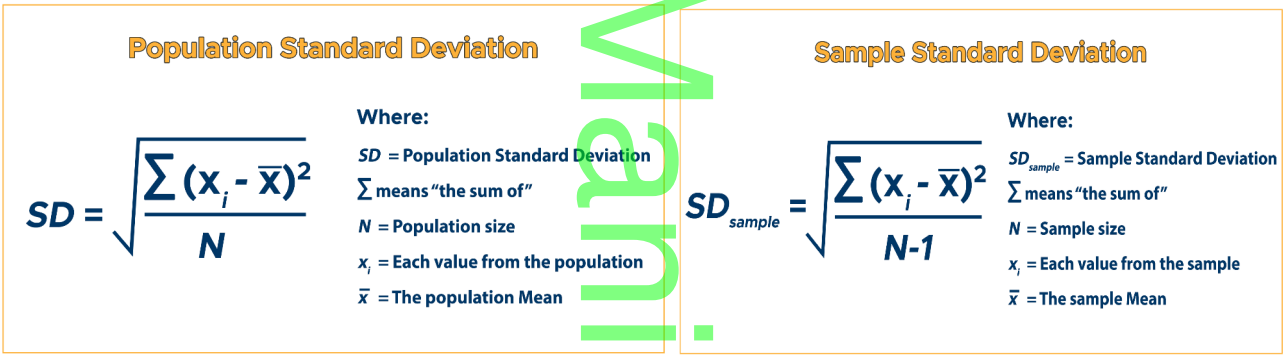
■ Measures of dispersion

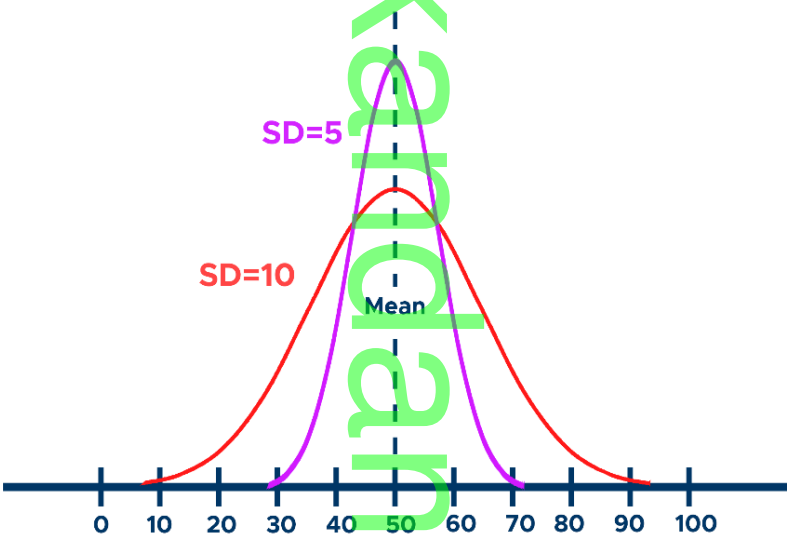
These describe the spread or variability of data points around the central value. Key measures include:

Range: The difference between the highest and lowest values. For {2, 3, 5, 8, 12}, the range is 10.

Variance: The average of squared differences from the mean, providing a detailed sense of spread.

Standard Deviation: The square root of the variance, giving a measure of spread in the same units as the data.





Compiled by Manikandan R

Interquartile Range (IQR): The difference between the 75th and 25th percentiles, showing the spread of the middle 50% of the data and resistant to outliers.

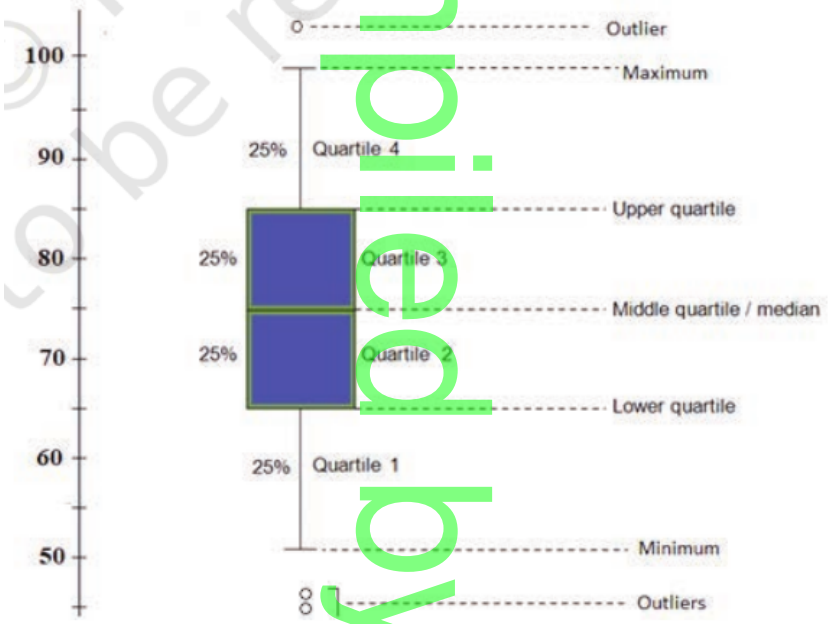
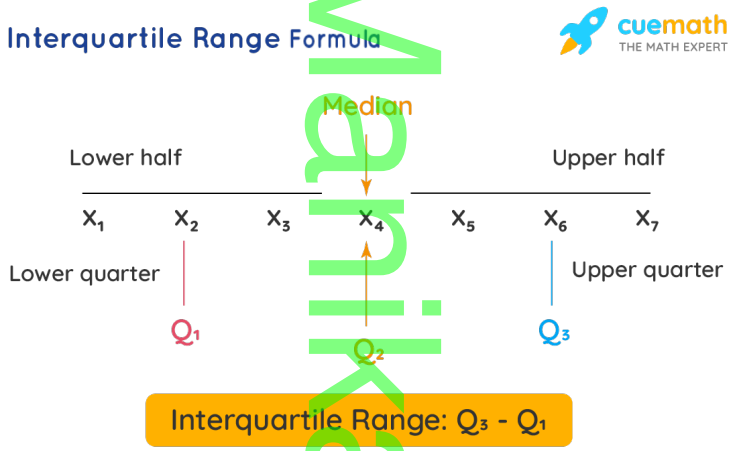
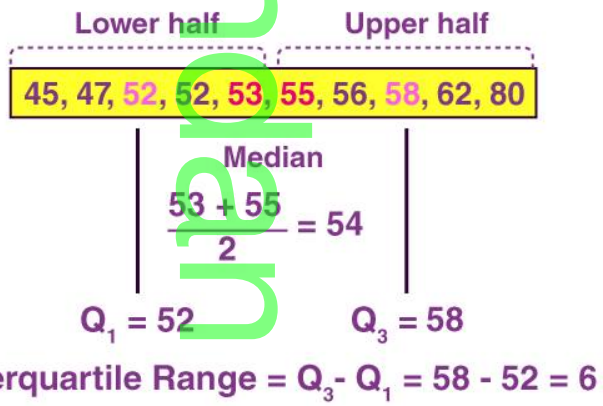


Figure 4.16: A Box Plot



Example



Compiled by Mathanan

6. Explain data retrieval and data preparation with examples.

Data retrieval is the process of locating, accessing, and extracting specific information from a storage system, such as a database, data warehouse, or cloud-based repository. Data preparation, often called data preprocessing or data wrangling, is the subsequent process of cleaning, transforming, and organizing that raw data into a usable format for analysis, reporting, and machine learning.

- **Data retrieval**

The goal of data retrieval is to obtain relevant data accurately and efficiently. The methods used vary depending on the type and location of the data.

Methods of data retrieval

Structured query language (SQL): This is used for retrieving data from relational databases where data is organized into tables with predefined relationships. Users write specific queries to filter, join, and sort data based on criteria.

Application programming interfaces (APIs): APIs define how software components should interact, allowing different systems to communicate and share data. They are commonly used to pull data from web services and cloud-based platforms.

Web scraping: This involves using automated tools to extract large amounts of data from websites. Web scrapers are particularly useful for gathering unstructured or semi-structured data.

Indexing and search engines: For large volumes of text-based data, search engines and indexing tools like Elasticsearch or Solr can quickly retrieve information by matching keywords or using more advanced natural language processing (NLP) techniques.

Caching: This method involves temporarily storing frequently accessed data in a high-speed storage location for quick retrieval, which reduces the need to access the original data source repeatedly.

- **Examples of data retrieval**

E-commerce product search: When a customer types a search query into an online store's website, the system retrieves relevant product information, including images, prices, and descriptions, from the product database.

Banking app: A user checking their account balance or transaction history is a simple but common example. The mobile application makes an API call to the bank's database to retrieve and display the requested financial data.

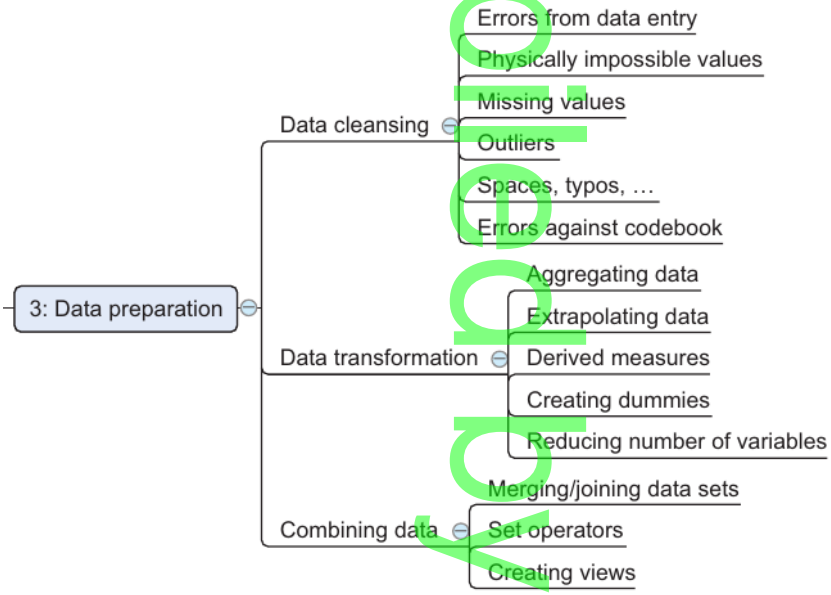
Healthcare patient records: A doctor or nurse accesses a patient's Electronic Health Record (EHR) system. The system retrieves the patient's medical history, lab results, and prescriptions from secure databases.

Compiled by Manikandan R

Social media feed: A social media platform retrieves real-time updates, photos, and personalized content by making API calls to different services and accessing its own data repositories.

- **Data preparation**

Data preparation is the time-consuming but critical phase that follows retrieval. It ensures that the collected data is high-quality, consistent, and correctly formatted for a chosen analytical model.



Steps Involved in data preparation

- **Data cleaning (Cleansing):** This step corrects or removes inaccurate, corrupt, incorrectly formatted, duplicate, or incomplete data.

Handling missing values: Decide how to treat missing data points. Strategies include

Deleting rows with missing values,

Filling them with a calculated mean or median, or

Using more advanced imputation models.

Removing duplicates: Identify and remove duplicate records that could otherwise bias the analysis.

Standardizing data formats: Ensure all entries in a field use the same format. For example, convert all dates to "YYYY-MM-DD" or all currency values to a single format.

Detecting and treating outliers: Identify and handle unusual data points that might skew results. Outliers might be corrected, removed, or kept, depending on whether they are considered true anomalies.

- **Data integration:** Combine datasets from multiple, disparate sources into a unified, coherent whole. This often involves merging data tables based on common fields.

Compiled by Manikandan R

- **Data transformation:** Convert data from one format or structure to another. This can involve both simple conversions and complex mathematical manipulations.

Feature engineering: Create new variables from existing ones to improve model performance. This requires domain knowledge and can be highly impactful.

Normalization/Standardization: Adjust numerical variables to a common scale to prevent features with larger values from dominating an algorithm.

Encoding categorical variables: Convert non-numerical, categorical data (like "red," "green," "blue") into a numerical format that machine learning models can process.

Data reduction: Reduce the volume of data while retaining its important information, which makes analysis more efficient.

Data validation: After preparation, check the data against predefined quality standards to confirm that all changes have been applied correctly and the data is ready for the next phase.

Example of data preparation

Imagine a retail company wants to analyze customer behavior to optimize marketing campaigns.
Data retrieval: The data team retrieves raw customer data from multiple systems:

CRM database: Customer names, contact info, and demographic details.

Online sales platform: Purchase history, including transaction IDs, product IDs, prices, and transaction dates.

Social media data: Customer engagement and comments scraped from public profiles.

Data preparation steps:

Cleaning: The team finds that some customer records have incomplete addresses or misspelled names. They also find and remove duplicate customer entries created by the sales platform. Date formats vary (some are "MM-DD-YY," others "DD-MM-YYYY"), so they are standardized.

Integration: The retrieved datasets are merged based on customer and product IDs to create a single, comprehensive dataset.

Transformation:

A new feature, `Customer_Age`, is created from the customer's birthdate.

The `Transaction_Date` is converted into a numerical format, and a new feature, `Purchase_Day_of_Week`, is engineered to capture weekly buying patterns.

Product prices are normalized to a 0-to-1 scale to give equal weight to other features in the analysis.

Rupendra Nandani Mandal

The `Gender` column (text: "Male," "Female") is encoded into numerical values (e.g., 0, 1) for the machine learning model.

Reduction: Irrelevant columns, such as a customer's internal ID from an old system, are removed.

Validation: The final dataset is checked to ensure that the newly created columns are accurate and that all transformations have been applied consistently.

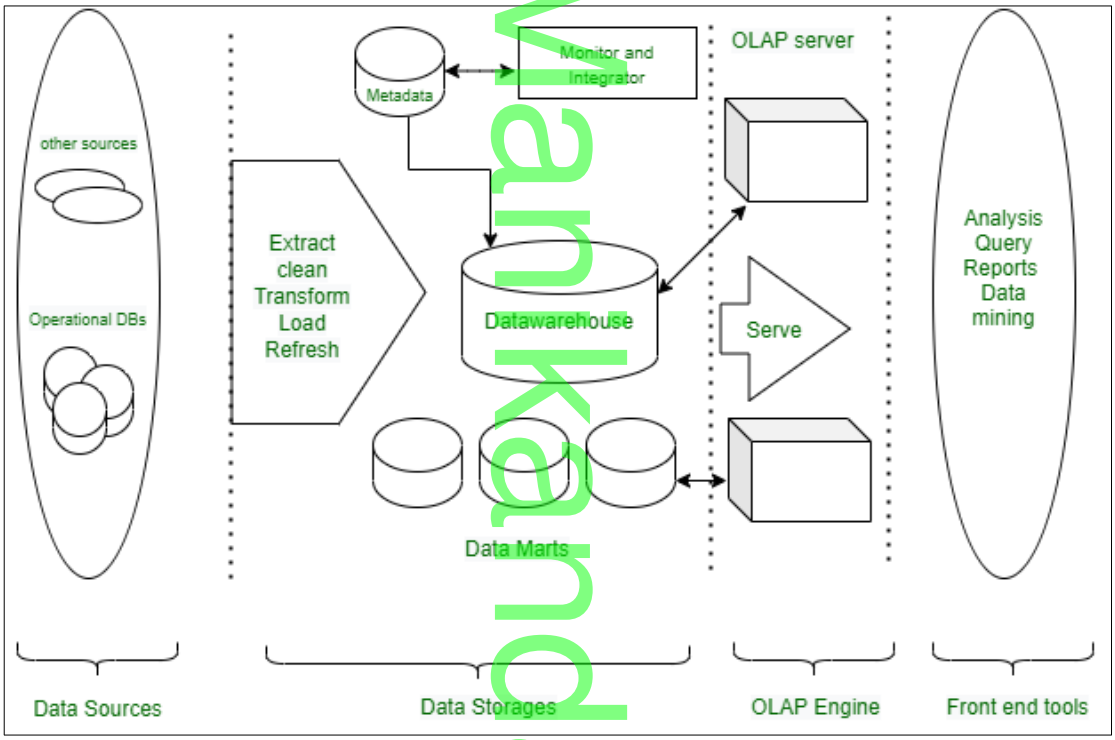
The resulting high-quality dataset is then ready for analysis, such as building a model to predict which customers are most likely to respond to a new marketing offer.

Data Warehousing

3 - Tier Architecture Data Warehouse

Data warehousing helps businesses make informed decisions using large datasets. The Three-Tier Architecture is widely used for its clear structure, dividing data processing into three layers for efficient access and management.

- Bottom Tier (Data Sources and Data Storage)
- Middle Tier (OLAP Engine)
- Top Tier (Front-End Tools)



3 - Tier Architecture Data Warehouse

1. Bottom Tier (Data Sources and Data Storage)

Bottom Tier is the foundation of the data warehouse, responsible for collecting, processing, and storing data from multiple sources. It plays a critical role in preparing data for analysis.

compiled by Nandkumar Rane

Key Components:

- **Data Sources:**
  Includes **operational databases (OLTP systems)**, **flat files**, **spreadsheets**, **external APIs**, **CRM/ERP systems**, and **web logs**. These are the raw inputs feeding into the data warehouse.
- **Data Storage:**
  Processed data is stored in a Relational Database Management System **(RDBMS)** or a **multidimensional database** designed to support structured querying and analysis.

2. ETL Process (Extract, Transform, Load)

ETL stands for Extract, Transform, and Load. It's a process where data is extracted from various sources, transformed for analysis, and loaded into a data warehouse.This is the core function of this bottom tier:

1. **Extract:**
   Gathers raw data from different, often incompatible sources.
2. **Transform:**
   Converts data into a consistent format, applying business rules, cleansing errors, handling missing values, and resolving duplicates.
3. **Load:**
   Loads the transformed data into the warehouse, organizing it for fast access and analysis.

This process ensures the warehouse contains clean, reliable, and business-ready data.

Common Challenges in Bottom Tier

Integrating data from diverse sources presents several challenges such as:

- **Data Quality:** Inconsistent data can lead to errors and unreliable analytics.
- **Data Compatibility:** Different data formats and structures can complicate integration.
- **Scalability:** Handling increasing volumes of data efficiently.

Solutions

- **Implement Robust ETL Tools:** Utilize powerful ETL tools like Informatica, Microsoft SSIS, or Confluent to streamline the data integration process.
- **Standardize Data Formats:** Standardizing data at the point of entry minimizes compatibility issues.
- **Continuous Data Quality Management:** Regularly check and clean data to maintain high quality.
- **Scalability Planning:** Design data storage solutions that can expand as data volume grows, ensuring that the architecture can handle future increases in data without performance degradation.

3. Middle Tier (OLAP Engine)

The **Middle Tier** hosts the **OLAP server**, which processes complex analytical queries. OLAP stands for Online Analytical Processing, which is a category of software tools that enables analysis of data stored in a database. Also, It acts as a bridge between the **data storage layer (bottom tier)** and the **user interface (top tier)**, ensuring data is quickly retrieved, aggregated, and ready for reporting and analysis.

Compiled by Manikandan R

OLAP is a powerful technology for complex calculations, trend analysis, and data modeling. It is designed for high-speed analytical processing. OLAP server models come in three different categories, including:

- **ROLAP (Relational OLAP):** This model uses a relational database to store and manage warehouse data. It is ideal for handling large data volumes as it operates directly on relational databases.
- **MOLAP (Multidimensional OLAP):** This model stores data in a multidimensional cube. The storage and retrieval processes are highly efficient, making MOLAP suitable for complex analytical queries that require aggregation.
- **HOLAP (Hybrid OLAP):** It is combination of relational and multidimensional online analytical processing paradigms. HOLAP is the ideal option for a seamless functional flow across the database systems when the repository houses both the relational database management system and the multidimensional database management system.

Common Challenges in Middle Tier

- **Data Latency:** Delays in data availability can impact decision-making.
- **Query Performance:** Managing large volumes of data can slow down query performance.
- **Data Integration:** Combining data from different sources with varying formats can be challenging.

Solutions

- **Real-Time & Incremental Loading:**
  Update data frequently using real-time and incremental loading to reduce latency and support faster decision-making.
- **Query Optimization:**
  Improve performance with **indexing**, **partitioning**, and **optimized SQL** for faster data retrieval.
- **Standardization & Integration Tools:**
  Standardize data formats and use tools like **Talend** or **Informatica** for seamless integration and improved data quality.

Top Tier

The Top Tier in the Three-Tier Data Warehouse Architecture comprises the front-end client layer, which is essential for interacting with the data stored and processed in the lower tiers. This layer includes a variety of ***business intelligence (BI) tools and techniques*** designed to facilitate ***easy access and manipulation of data for reporting, analysis, and decision-making.***

BI tools are critical components of the Top Tier, providing robust platforms through which users can query, report, and analyze data. Popular BI tools include:

- **IBM Cognos:** Offers comprehensive reporting capabilities.
- **Microsoft BI Platform:** Integrates well with existing Microsoft products, providing a familiar interface for users.
- **SAP BW:** Specializes in managing large datasets and integrating with other SAP products.
- **Crystal Reports:** Known for its powerful reporting features.
- **SAS Business Intelligence:** Provides advanced analytics.
- **Pentaho:** A versatile tool for data integration and visualization.

The Top Tier is crucial for decision-making as it provides the interface through which insights are accessed and explored. By presenting data in visual formats such as graphs, charts, and dashboards,

Prepared by Manikanda

these tools allow decision-makers to quickly grasp complex patterns, trends, and anomalies, leading to faster and more effective decision-making.

Common Challenges in Top Tier

- **Usability Issues:** Complex tools can hinder user adoption and effectiveness.
- **Integration Difficulties:** Ensuring seamless integration with other tiers can be challenging.

Solutions

- **User Training and Support:** Offering comprehensive training sessions to help users fully leverage the capabilities of BI tools.
- **Choosing Integrative Tools:** Select tools that easily integrate with existing systems in the data warehouse architecture, ensuring consistency and reliability in data handling.

Excat Clear Distinction Between Data Science And Data Mining And Data Warehouse And Data Warehousing With Authenticated Source ?

Data Science

Data Science is an interdisciplinary field that uses scientific methods, algorithms, statistical techniques, and systems to extract insights and knowledge from structured and unstructured data. It covers the full data lifecycle, including data collection, cleaning, analysis, modeling, and the generation of actionable insights and predictions. *Data Science is broader than data mining and subsumes it as one of its analytical techniques*.

Data Mining

Data Mining is a specialized subset of data science. It focuses explicitly on uncovering previously unknown patterns, relationships, or anomalies within large datasets. Its core purpose is knowledge discovery, often employing algorithms and statistical methods to extract useful information from existing data. It is narrower in scope than data science and typically works on “mining” already collected and stored data.

Data Warehouse

A **Data Warehouse** is a central, structured repository designed to store and manage large volumes of data collected from various sources. It is specifically engineered for reporting and data analysis to support business decision-making. Data warehouses store historical data, enabling organizations to query and analyze information over long periods.

Data Warehousing

Data Warehousing is the process or set of activities related to designing, building, and managing data warehouses. It covers all operations required for data extraction, transformation, loading (ETL), maintenance, and ensuring data quality and consistency.

In summary, *data warehousing* refers to the technical and strategic process, while *data warehouse* is the physical system or structure itself.

Compiled by Mabikandan R

Key Differences / Distinctions

Term	Definition	Main Purpose	Scope
Data Science	Broad field employing scientific and computational methods to extract insights from data	Insight generation, prediction, modeling	End-to-end data lifecycle
Data Mining	Subset focusing on finding unknown patterns and relationships in large datasets	Knowledge discovery, pattern extraction	Analytical, post-collection
Data Warehouse	Centralized repository for storing structured, historical data	Storage and ready access for analysis	Physical, architectural
Data Warehousing	Processes involved in building, maintaining, and managing data warehouses	Data integration, ETL, organized storage	Technical, procedural

Data Warehouse Workflow

- **Objective:** Store, consolidate, and organize large volumes of data from diverse sources for efficient querying and reporting.

Key Steps:

- **Data Extraction:** Gather data from multiple, often heterogeneous, sources.
- **Data Cleaning and Transformation:** Ensure consistency, resolve duplicates, remove noise, and standardize formats.
- **Data Loading:** Move processed data into a centralized repository using ETL (Extract, Transform, Load) processes.
- **Schema Design:** Structure the data to support business analytics (e.g., star or snowflake schemas).
- **Maintenance and Updates:** Periodically refresh data and maintain integrity.
- **Reporting and Querying:** Make cleaned, historical data available for analysis and business intelligence.

Data Mining Workflow

- **Objective:** Discover patterns, trends, and actionable insights from existing data, often already stored in a data warehouse.

Key Steps:

- **Data Selection:** Choose relevant subsets of data for analysis, usually from data warehouses or other stores.
- **Data Preprocessing:** Prepare the data by handling missing values, normalizing, and reducing dimensionality.
- **Model Building:** Apply statistical, machine learning, or AI algorithms to find hidden patterns.

compiled by Vamikandan R

- **Pattern Evaluation:** Assess and interpret discovered knowledge for significance and actionable value.
- **Visualization & Deployment:** Present findings through reports or dashboards; integrate insights into decision-making processes.

1. **Iterative Refinement:** Iterate over different models and parameters to improve accuracy and relevance.

Comparison Table: Data Warehouse vs Data Mining Workflow

Aspect	Data Warehouse Workflow	Data Mining Workflow
Purpose	Data consolidation, organization, and storage	Pattern discovery and knowledge extraction
Primary Tasks	Extraction, cleaning, transformation, loading, schema design, maintenance	Selection, preprocessing, modeling, evaluation, visualization
End Users	Engineers, database admins, BI analysts	Data analysts, data scientists, business users
Outcome	Reliable, historical data repository for analytics	Actionable insights, trends, and predictions
Dependency	Forms the foundation for mining, must precede data mining	Relies on clean, consolidated data (often from a warehouse)

Which tools and roles are essential during a warehousing ETL phase

During the ETL (Extract, Transform, Load) phase of data warehousing, specific tools and roles are essential to ensure high data quality, efficient integration, and smooth operation.

Essential ETL Tools

- **Apache NiFi:** Open-source tool for real-time, automated data flow management across systems.
- **Talend:** Open-source ETL suite supporting batch and real-time integration of large data volumes.
- **Microsoft SSIS (SQL Server Integration Services):** Commercial platform for scalable, automated data extraction, transformation, and loading.
- **Hevo Data:** Modern, cloud-based ETL platform automating data integration and supporting real-time pipelines.
- **Oracle Warehouse Builder:** ETL and data integration platform for large-scale data management in Oracle environments.
- **TROCCO:** Fully managed cloud ETL solution offering automation, security, and advanced data transformation features.
- **Others:** Tools like Informatica PowerCenter, AWS Glue, SAP Data Services, and Google Dataflow are also widely used depending on integration requirements and business needs.

Essential Roles

- **ETL Developer:** Designs and implements the ETL workflows and pipelines, translating business requirements into technical solutions.

Downloaded by Wanikandan R

- **Data Engineer:** Builds and manages infrastructure, optimizes performance, and ensures scalability and reliability of ETL pipelines.
- **Data Architect:** Develops overarching architecture, including source system mapping, schema design, and data modeling for the warehouse.
- **Database Administrator (DBA):** Ensures integrity, security, backup, and optimal performance of the data warehouse.
- **Quality Assurance (QA) Engineer/Tester:** Verifies ETL workflows meet data quality standards, validates transformation logic, and ensures data accuracy and consistency.
- **Project Manager:** Plans and coordinates the ETL team's work, aligning it with stakeholder expectations and business schedules.
- **Business Analyst:** Provides requirements, interprets ETL results, and ensures business needs are met by the data integration process.

Summary Table of Tools and Roles

Category	Examples (Tools/Roles)	Key Responsibilities
ETL Tools	Apache NiFi, Talend, MS SSIS, Hevo, Oracle WB, TROCCO, Informatica	Automate extract, transform, load processes
ETL Developer	N/A	Design, develop, and test ETL workflows
Data Engineer	N/A	Build infrastructure, optimize pipelines
Data Architect	N/A	Data modeling, mapping, system design
DBA	N/A	Manage warehouse security and performance
QA Engineer	N/A	Validate data quality and ETL correctness
Project Manager	N/A	Coordinate teams and manage timelines
Business Analyst	N/A	Specify requirements and review outcomes

How to comprehend Descriptive and Inferential Statistics

STATISTICS
Population & Sample

In statistics, a population refers to any complete collection of observations or potential observations, whereas a sample refers to any smaller collection of actual observations drawn from a population.

Descriptive Statistics

Statistics which provides us the tools tables, graphs, averages, ranges, correlations for organizing and summarizing the inevitable variability in collections of actual observations or scores.

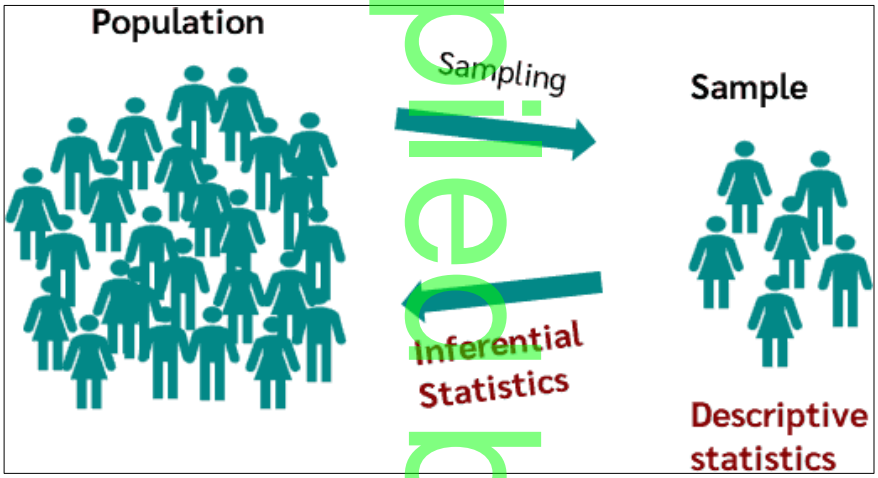
Examples are:

1. A graph showing the annual change in global temperature during the last 30 years
2. A report that describes the average difference in grade point average (GPA) between college students who regularly revised subjects and those who don't.

Compiled by Manikandan R

Inferential Statistics

Inferential statistics is concerned with generalizing beyond sets of actual observations, that is, with generalizing from a sample to a population. Also it provides tools for a variety of tests and estimates for generalizing beyond collections of actual observations



DESCRIPTIVE STATISTICS

Vs

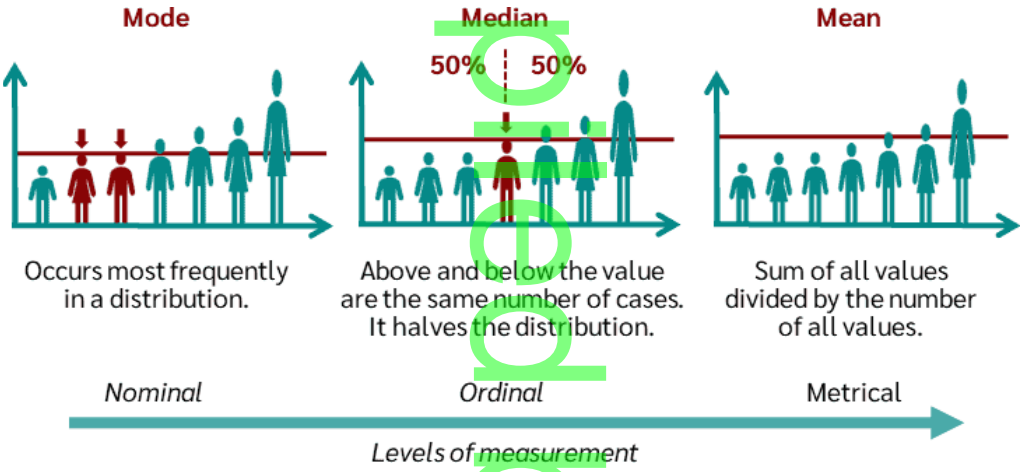
INFERENTIAL STATISTICS

DESCRIPTIVE	INFERENTIAL
It is the analysis of data that helps to describe, show and summarize data under study	It is the analysis of random sample of data taken from a population to describe and make inference about the population
Organize, analyze and present data in a meaningful way	Compares, test and predicts data
It is used to describe a situation	It is used to explain the chance of occurrence of an event
It explain already known data and limited to a sample or population having small size	It attempts to reach the conclusion about the population
Types: Measure of central tendency & Measure of variability	Types: Estimation of parameters & Testing of hypothesis
Results are shown with help of charts, graphs, tables etc.	Results are shown with help of probability scores

Compiled by Manikandan R

Illustrate with an example of Parameters for center location or Measure of Central Tendency

In descriptive statistics, mean, median and mode are location parameter (measures of central tendency). Based on data collected in a sample, the location parameter provide information about where the "center" of the distribution lies.



Measures of location can be used to summarize or describe a list of data with only one parameter. An example would be that the average duration of studies of sports students at the university XY is 11.1 semesters.

Mean (arithmetic mean)

Definition:

The **arithmetic mean** is the sum of all observations divided by their number *n*.

The mean value can only be calculated for **metric variables**, i.e. if a metric scale of measurement is given. It indicates where the center of gravity of a distribution can be found. In everyday life it is also called the "average".

The mean value can be calculated by adding up all the values of a variable and then dividing the sum by the number of characteristic values.

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Mean Value — Number of values — Value at the i-th position

Median

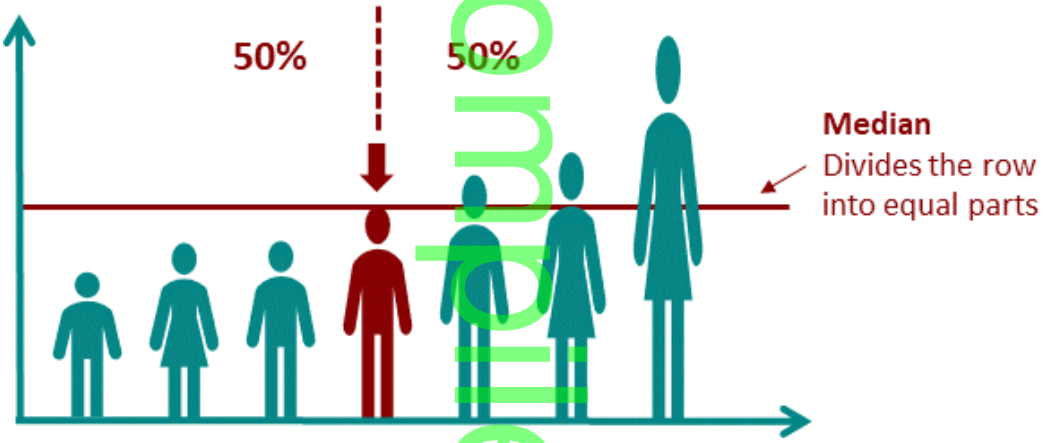
Definition:

In an ordered series, the **median** is the value that divides the series into an equal upper and lower range.

If the measured values of a variable are ordered by size, the value in the middle is the median. The median is therefore the "middle value" of a distribution. It leads to a division of the series into two parts: one half is smaller and one is larger than the median.

Since for the calculation of the median the data are ordered, the variables must have ordinal or metric scale level.

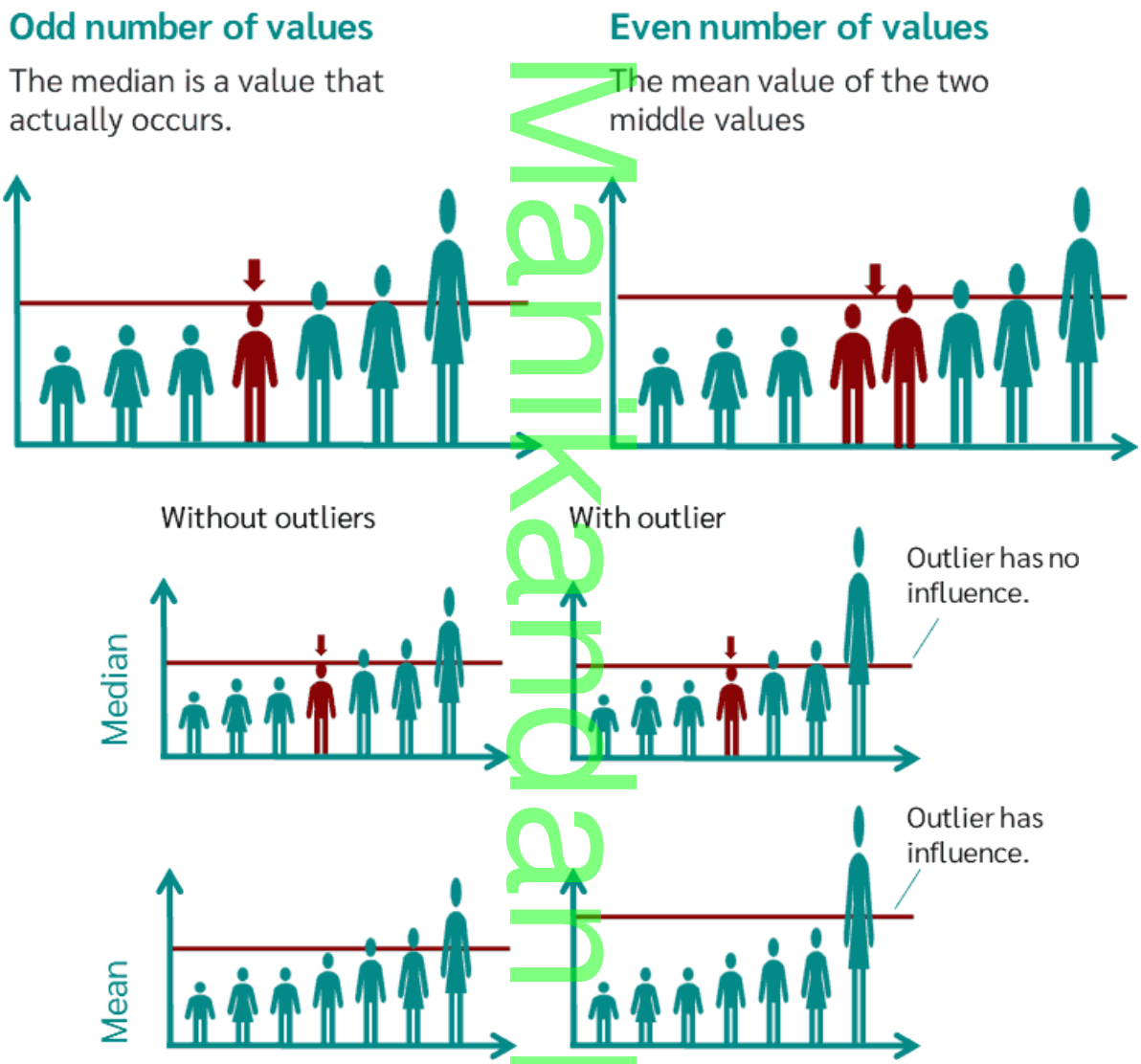
Prepared by Mr. Nikhil Mandal



For the **median** to be calculated, the variable must be **ordinally scaled**. Ordinal scaling means that there is a **ranking** order between the values of a variable. This applies, for example, to school grades, height or salary. However, it is not possible to create a ranking for a variable Place of birth and therefore the median cannot be calculated here.

If there is an **odd number** of characteristic values, then the median is a value that actually occurs.

If there is an **even number** of characteristic carriers (persons), the two middle characteristics are added together and their sum is divided by two.



Compiled by Manikandan R

Mean vs Median

Compared to the mean, the median is much more robust against scattering. An outlier usually has no influence on the median, but it has a more or less large influence on the mean.

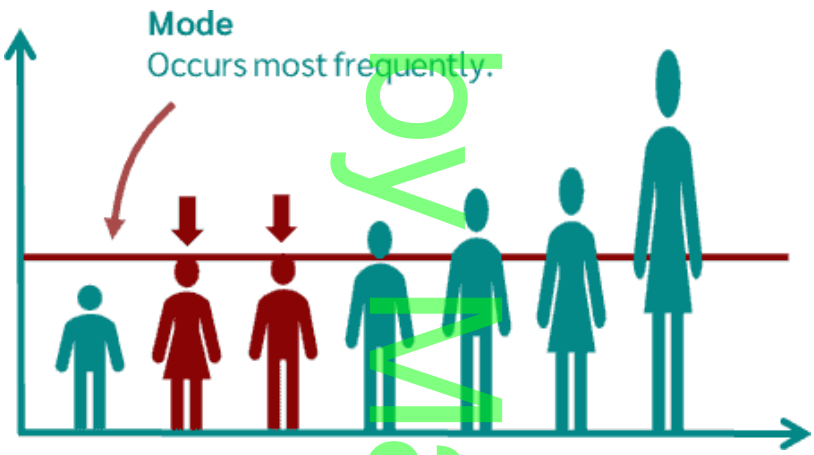
Mode (Modal Value)

The mode is the most common value. The mode is therefore the most frequent value in a distribution and corresponds to the highest value in the distribution. It is therefore the value that is "typical" for a distribution.

The mode can be used for both **metric** and **categorical** (nominal or ordinal) variables.

Definition:

The **mode** is the value of a distribution that occurs most often.



Advantage and disadvantage of the Mean, Median and Mode

If the distribution is symmetric, the mean and median are equal, and if the distribution is symmetric and unimodal, all three measures are equal. As a rule, however, the three measures have different values. Now, of course, the question is which of the measures of central tendency to use. Unfortunately, there is no clear rule for this, only a few decision aids.

Mean: The mean value is by far the most used. The disadvantages of the mean are that it is sensitive to outliers, the value does not have to exist in the data and for the interpretation to be meaningful, the data should have metric scale level.

Median: The great advantage of the median is that it is very robust against outliers and that the data only have to be scaled ordinally.

Mode: The mode is the value that occurs most frequently, which has the advantage that the value actually occurs. Furthermore, the mode can also be calculated for data that cannot be ordered and thus have a nominal scale level. The disadvantage is that the mode does not take into account the other existing data.

Frequency Distribution

A frequency distribution is a collection of observations produced by sorting observations into classes and showing their frequency (f) of occurrence in each class.

Compiled by Wanbo Kanan

Ungrouped data

When observations are sorted into ***classes of single values,*** as in Table, the result is referred to as a frequency distribution for ungrouped data.

Grouped data frequency Distribution

When observations are ***sorted into classes of more than one value***, as in Table 2.2, the result is referred to as a frequency distribution for grouped data. Let's look at the general structure of this frequency distribution.

Relative Frequency

Relative frequency distributions show the frequency of each class as a part or fraction of the total frequency for the entire distribution.

Cumulative Frequency Distribution

A frequency distribution showing the total number of observations in each class and all lower-ranked classes.

Compiled by Manikandan R